Perturbed Public Voices (P^{2}V): A Dataset for Robust Audio Deepfake Detection

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Abstract

Current audio deepfake detectors cannot be trusted. While they excel on controlled benchmarks, they fail when tested in the real world. We introduce Perturbed Public Voices (P^{2}V), an IRB-approved dataset capturing three critical aspects of malicious deepfakes: (1) identity-consistent transcripts via LLMs, (2) environmental and adversarial noise, and (3) state-of-the-art voice cloning (2020–2025). Experiments reveal vulnerabilities of 22 recent audio deepfake detectors: models trained on current datasets lose 43% performance when tested on P^{2}V, with performance measured as the mean of F1 score on deepfake audio, AUC, and 1-EER. Simple adversarial perturbations induce up to 16% performance degradation, while advanced cloning techniques reduce detectability by 20-30%. In contrast, P^{2}V-trained models maintain robustness against these attacks while generalizing to existing datasets, establishing a new benchmark for robust audio deepfake detection. P^{2}V is publicly available: https://sites.northwestern.edu/nsail/p2v-perturbed-public-voices/.

CCS Concepts

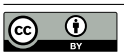
• **Security and privacy → Social aspects of security and privacy**; • **Computing methodologies → Machine learning approaches**;

Keywords

Audio Deepfake Detection, Voice Cloning, Synthetic Speech, Robustness Evaluation, Benchmark Dataset

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1 Introduction

Rapid advances in generative AI have ushered in an era of hyper-realistic synthetic media, with audio deepfakes emerging as a powerful tool for misinformation [8, 56]. From impersonating public figures in fraudulent speeches[1] to manipulating financial markets with fabricated CEO statements[2], malicious applications of this technology threaten democratic discourse, security, and trust in digital content [31, 38]. While researchers have developed benchmark datasets and detectors [46, 51, 53], existing resources fail to capture the complexity of real-world deepfakes [24, 41], where synthetic audio is often embedded in noisy environments, semantically aligned with a speaker's identity, and generated using SOTA voice cloning methods.

Current datasets, such as those derived from the ASVspoof Challenge series [18, 19, 37, 39, 43, 46], prioritize controlled laboratory conditions, omitting critical variables like environmental noise and contextual plausibility. Recent efforts like In-The-Wild [24] and DEEP-VOICE [3] compile online deepfakes of celebrities, but have a homogeneous set of speakers and lack generalization ability as they rely on easily detected deepfakes that lack the sophistication of real-world malicious deployments. This mismatch between training data and real-world conditions creates a false sense of security: *detection models achieve near-perfect performance on benchmark tasks but fail when confronted with the nuanced, adversarially perturbed deepfakes in the real world.*

We present Perturbed Public Voices (P^{2}V), a large-scale, IRB-approved[3] dataset that bridges this gap. It consists of 257,440 samples from 206 public figures. By systematically integrating transcripts tailored to a speaker's public persona using 3 different Large Language Models (LLMs), acoustic variations with 10 perturbations simulating natural and adversarial distortions, and 10 state-of-the-art text-to-speech/voice cloning and voice conversion methods (spanning 2020-2025), we enable the development of detection systems robust to emerging threats.

Our comprehensive experiments, involving 22 state-of-the-art audio deepfake detection methods, reveal stark limitations in current approaches: models trained on existing benchmarks show dramatic performance drops when tested on P^{2}V—e.g., deepfake

[1] https://www.npr.org/2024/05/30/nx-s1-4986088/deepfake-audio-elections-politics-ai
[2] https://blackbird.ai/blog/celebrity-deepfake-narrative-attacks
[3] IRB approval number: STU00223255

detection scores, measured as the mean of F1 scores on the fake class, AUC and 1-EER, decaying by 43%—, while P^{2}V-trained models generalize effectively to in-the-wild data. Additionally, simple adversarial perturbations (e.g., background noise and Gaussian noise injection) can cause up to 16% performance degradation of deepfake detectors. Beyond benchmarking, P^{2}V provides a framework for evaluating the detectability of novel synthesis techniques. We demonstrate that newer voice cloning methods (e.g., XTTS-v2 [5] and IndexTTS [12]) are 20–30% harder to detect than older approaches. By open-sourcing P^{2}V, we aim to catalyze progress toward the development of more robust detection systems for real-world deployment.

2 Related Work

Early efforts to establish publicly available benchmarks for audio deepfake detection, such as the ASVspoof Challenge series [19, 37, 39, 46], established standard protocols for evaluating deepfake detectors. These datasets, grounded in laboratory conditions, pair clean speech recordings with synthetic audio produced via classical text-to-speech (TTS) and voice conversion (VC) models. While they support algorithmic benchmarking, they omit key factors present in real-world settings, including environmental noise, adversarial perturbations, and identity-context alignment.

Subsequent initiatives sought to increase ecological validity. The Audio Deep Synthesis Detection (ADD) Challenge [49, 50] introduced background noise and music, while other datasets embedded short fake segments within real utterances [54, 55]. Recent datasets have extended scope along multiple axes: multilinguality (e.g., WaveFake [14], CFAD [23], MLAAD [25]), emotional manipulation (e.g., EmoFake [58]), domain-specific scenarios (e.g., Fake Song Detection [45]), or detailed method annotations (e.g., SFR [47], Codecfake [44]). TIMIT-TTS [33] incorporates visual cues for multimodal detection, while CLAD [41] explores semantic editing.

The In-The-Wild (ITW) dataset [24] compiles deepfakes of 58 public figures sourced from online platforms. However, many samples are easily identified as fake, as they feature well-known personalities speaking absurd or fictional content (e.g., Donald Trump reads Star Wars). As the authors note, these clips are suitable for evaluation, but inadequate for training, due to their semantic implausibility and limited variability. Similarly, the DEEP-VOICE dataset [3] focuses on celebrities but lacks consistent annotation and perturbation modeling.

P^{2}V addresses key limitations in existing corpora by systematically varying both synthesis and acoustic conditions. First, it incorporates deepfakes generated using ten state-of-the-art TTS and VC models spanning varied architectures and training paradigms, enabling a controlled evaluation of how different generative techniques impact detection difficulty. Second, each deepfake is based on semantically aligned transcripts derived from real public discourse by well-known figures, ensuring contextual plausibility and speaker consistency. Third, to emulate real-world deepfakes, we introduce ten perturbations (e.g., Gaussian noise, background noise, air absorption) that reflect both environmental variability and potential adversarial interference. Audio samples are annotated with the corresponding generation method and perturbation, allowing for fine-grained attribution of detection performance across synthesis pipelines and noise conditions.

3 Perturbed Public Voices (P^{2}V)

This section provides a detailed description of how we built our dataset. The overview of the construction workflow is illustrated in Figure 1.

3.1 Reference Audio Collection

Selection of Deceased Public Figures (DPFs). We restricted our dataset to DPFs for ethical and legal reasons. Generating realistic deepfake audio of living individuals, even for research purposes, raises significant concerns: it could facilitate misuse if samples were extracted from the dataset, potentially harm reputations, and may violate personality rights in various jurisdictions. By focusing exclusively on DPFs, we mitigate these risks while still enabling research on the voices of prominent figures whose speech patterns, public personas, and media presence closely mirror those of living targets that malicious actors typically impersonate. This approach received IRB approval (STU00223255) and balances the need for realistic, high-stakes evaluation data against the ethical imperative to avoid contributing to potential harms against living individuals.

We identified DPFs using WikiData, with data collection spanning from January 1, 2012, to May 1, 2025. We focused on individuals from professions frequently targeted by deepfake manipulation: politicians, journalists, actors, musicians, writers, television presenters, film directors, philosophers, and activists. Manual curation verified the availability of high-quality audio samples for 206 public figures.

Publicly-Available Audio Retrieval. To simulate realistic conditions under which malicious actors might operate (i.e., accessing public sources to obtain reference speech for voice cloning of public figures), we collected authentic audio samples from YouTube. Our search methodology used keywords specifically targeting interviews, speeches, panel discussions, podcasts, and other public speaking engagements featuring the identified public figures. We prioritized high-quality recordings with minimal background noise, clear articulation, and extended duration to ensure sufficient speech material for robust voice analysis. This approach mirrors the practical constraints and opportunities that would confront those attempting unauthorized voice replication, thereby enhancing the ecological validity of our dataset.

Speech Diarization and Speaker Identification. Since the collected audio samples for the 206 DPFs contain multiple speakers, we isolate the target DPFs' speech content. We used Pyannote [4, 28], a state-of-the-art speech diarization framework that segments audio into discrete units (corresponding to individual utterances) and assigns unique speaker IDs to each segment. Following diarization, we implemented a post-processing step where consecutive segments with identical speaker identifiers were merged into single contiguous segments, with a maximum duration of 40 seconds. This automated process was also manually verified to accurately associate speaker IDs with either the target DPF or an *unknown* label. To minimize privacy concerns, segments labeled *unknown*

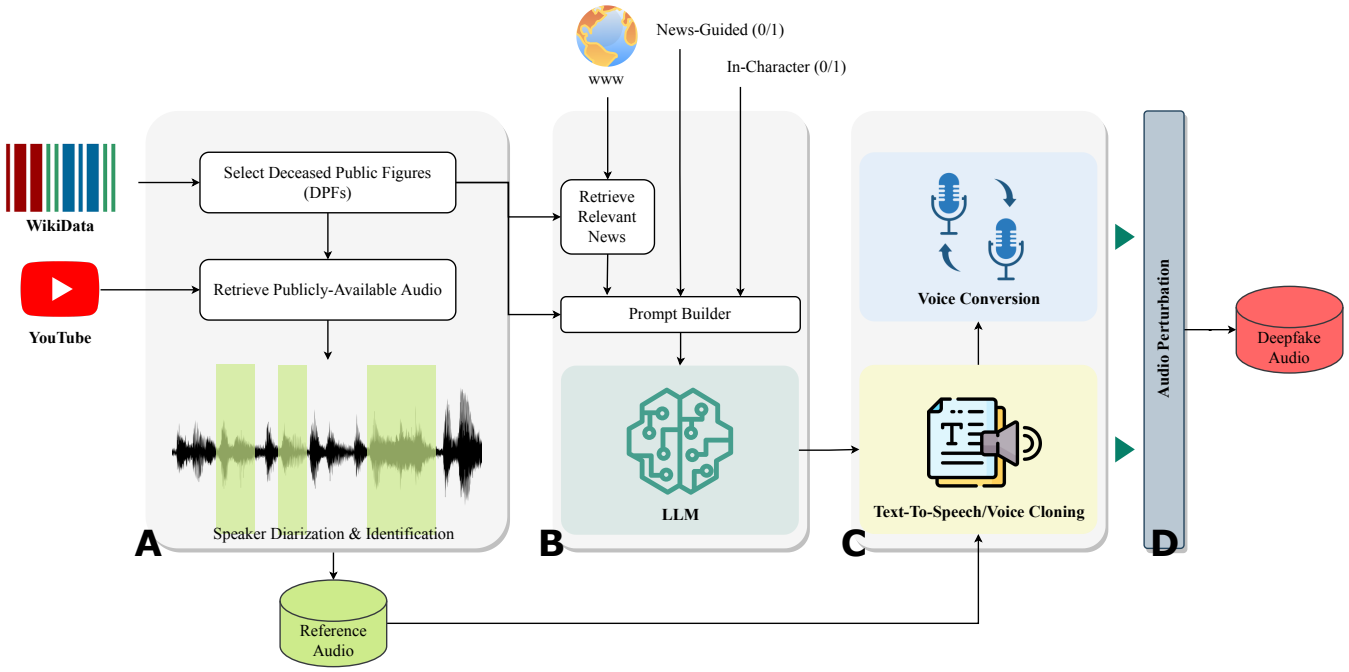


Figure 1: Overview of the audio deepfake generation pipeline for the P^{2}V dataset. The workflow consists of four main stages: (A) *Reference audio collection* from WikiData and YouTube, followed by speaker diarization and identification to isolate target speaker segments; (B) *Fake transcript generation* using Large Language Models (LLMs) with news-guided and in-character/out-of-character conditioning; (C) *Audio deepfake synthesis* through Text-to-Speech (TTS) models and Voice Conversion (VC) methods; and (D) *Audio perturbation application* to simulate real-world acoustic conditions and adversarial distortions, ultimately producing the final deepfake audio dataset.

were excluded from the dataset. For quality control and consistency, we also excluded segments shorter than 5 seconds (which typically contain insufficient reference content for voice cloning methods) and longer than 60 seconds (which may include unwanted artifacts or speaker transitions due to diarization errors).

3.2 Fake Speech Transcript Generation

To simulate a broad spectrum of real-world scenarios in which synthetic audio might be deployed, we enriched our dataset with temporally grounded transcripts associated with each DPF. Specifically, we leveraged the date of death for each individual to construct a timeline of potentially relevant public discourse. For each DPF, we selected 10 random distinct articles published within the five years preceding their death. These articles were identified through targeted search queries designed to retrieve publicly available content about the DPF at various points in time.

LLMs are increasingly demonstrating impressive capabilities in role-playing tasks [22, 34], including a nuanced understanding of public figures and celebrities [7, 9, 52]. Leveraging this strength, we generate four distinct categories of transcripts by combining in-character versus out-of-character speech patterns and news-guided versus non-news-guided content.

We prioritized LLMs that balance performance and computational efficiency, specifically those with 10 to 30 billion parameters. To ensure transparency and reproducibility, we use open-source models rather than proprietary ones such as ChatGPT or Gemini. We selected three instruction-tuned models from established open-source LLM families that consistently demonstrated strong instruction adherence and formatting compliance [26, 40]: *Gemma-3* [36] with 27B parameters, *Qwen2.5* [48] with 14B parameters, and *Mistral* [4] with 24B parameters. To ensure consistency in audio duration (approximately 30 seconds), each LLM was prompted to generate transcripts containing around 100 words. For each model, we use the same prompts (see Appendix).

3.3 Audio Deepfake Generation

To generate high-fidelity audio deepfakes, we combined our curated transcripts with public reference audio using ten state-of-the-art generative pipelines, comprising seven text-to-speech (TTS) models and three voice conversion (VC) methods. Each TTS model synthesizes speech conditioned on a semantically aligned transcript and a randomly selected target-speaker reference. We employ several architectures: Tacotron DCA [2], YourTTS [6], XTTS [5], VEVO [57], Zonos [5], CosyVoice 2 [13], and IndexTTS [12], covering a range of capabilities from multilingual and zero-shot synthesis to fine-grained control over prosody and emotional expression. To

[4] https://huggingface.co/mistralai/Mistral-Small-24B-Instruct-2501
[5] https://github.com/Zyphra/Zonos

increase speaker variability and evaluate layered synthesis effects, we also adopt a two-stage pipeline in which speech generated by IndexTTS is transformed via voice conversion using Diff-HierVC [10], FreeVC [20], or Seed-VC [21]. These models enable text-free, style-controllable, and speaker-adaptive conversion in zero-shot settings. All methods use officially released pretrained weights to ensure reproducibility. Each audio sample is labeled with its generative method, facilitating attribution in downstream experiments.

3.4 Audio Perturbations

To simulate real-world acoustic variability and assess model robustness, we apply one of ten perturbations to each generated audio deepfake, selected uniformly at random, with associated hyperparameters sampled per transformation. Each sample is labeled with its perturbation type and parameters to ensure full reproducibility.

The ten perturbations are: (1) *Background noise addition*: one audio file is sampled from the ESC-50 dataset [27], which includes 2,000 labeled environmental sounds. (2) *Gaussian noise injection*: noise amplitude is fixed at a minimum of 0.001 and sampled from 0.005, 0.01, 0.015, 0.02 for the maximum. (3) *Air absorption simulation*: source–microphone distance is selected from 10, 20, 50, 100 meters to replicate frequency-dependent attenuation over distance. (4) *MP3 compression*: lossy encoding is applied with bitrates drawn from 8, 16, 32, 64 kbps to reflect online transmission degradation. (5) *Pitch shifting*: semitone offsets are sampled from ±2, ±4, ±6, and ±12 to test resilience to pitch-based manipulation. (6) *Band-pass filtering*: frequency ranges are chosen from (200, 4000), (150, 5000), or (50, 8000) Hz, simulating telephone-quality or bandwidth-limited audio. (7) *Time stretching*: playback speed is modified by a factor selected from 0.6, 0.8, 1.2, 1.4, preserving pitch but altering rhythm and cadence. (8) *Audio reversal*: the waveform is reversed in time, eliminating temporal structure without requiring parameter tuning. (9) *Impulse response convolution*: reverberation is added using one audio sample randomly selected from the EchoThief Impulse Response Library[6]. (10) *Clipping distortion*: amplitude saturation is introduced using percentile thresholds drawn from (0, 20), (10, 40), (20, 60), controlling distortion severity. These perturbations form a controlled yet varied augmentation suite designed to reflect environmental noise, adversarial conditions, and common signal degradations, thereby supporting rigorous and ecologically valid evaluation of audio deepfake detection systems.

3.5 Dataset Statistics and Organization

Table 1 provides a breakdown of the dataset, illustrating the systematic organization of samples across different generation methods and data splits. The dataset contains 257,440 audio samples derived from 206 unique subjects, partitioned into training (131 subjects), validation (37 subjects), and test (38 subjects) sets to ensure robust model evaluation and prevent data leakage across speaker identities. The dataset follows standard practices [35], with approximately 60% of samples allocated to training, 20% to validation, and 20% to testing. The subject-level partitioning guarantees that no speaker appears across multiple splits, thereby preventing overfitting to specific vocal characteristics and promoting generalization to unseen speakers. Like most existing benchmarks (e.g. ASVspoof [37], WaveFake [14], CodecFake [44]), P^{2}V contains a disproportionately high number of fakes, in contrast to current real-world scenarios where real speech prevails. To address potential biases, we report class-specific metrics in our experiments and encourage future research to carefully take this imbalance into account.

[6]https://www.echothief.com/

Table 1: P^{2}V distribution of subjects and audio samples across training, validation, and test splits. The dataset contains both synthetic (fake) audio generated through various methods and authentic (real) audio samples.

Category	Training	Validation	Test	Total
Subjects	131	37	38	206
Synthetic Audio Samples				
Samples per LLM (3)	52,400	14,800	15,200	82,400
Samples per Voice Cloning Method (10)	15,720	4,440	4,560	24,720
Samples per Noise Perturbation (10)	15,720	4,440	4,560	24,720
Fake Audios (Total)	**157,200**	**44,400**	**45,600**	**247,200**
Real Audios	**6,237**	**1,963**	**2,040**	**10,240**
Grand Total	**163,437**	**46,363**	**47,640**	**257,440**

To facilitate evaluation under varying acoustic conditions, we provide both a clean, original version of the dataset P^{2}V[o], and a perturbed version P^{2}V[p] incorporating noise perturbations.

4 Experiments

Our experiments examine four key dimensions: (1) cross-dataset generalization performance comparing our dataset against existing benchmarks (InTheWild [24]), (2) the impact of state-of-the-art voice cloning technologies incorporated in our dataset on detection accuracy, (3) the influence of varied transcript generation methods on detection systems, and (4) vulnerability to common audio perturbations that adversaries might employ to evade detection. These analyses reveal significant limitations in current detectors and validate the utility of our dataset.

4.1 Experimental Setup

4.1.1 Dataset. We demonstrate the robustness and generalization ability of P^{2}V against a related real-world dataset, In-The-Wild (ITW) [24]. The ITW dataset contains deepfakes of 58 public figures extracted from publicly available video and audio content, containing 19,963 authentic and 11,816 deepfake samples. Similarly to P^{2}V, ITW was partitioned into training, validation and test sets using stratified sampling to preserve the original class distribution.

The ITW dataset consists of speakers engaging in absurd and out-of-character dialogue, which may introduce biases that could limit model generalization when trained exclusively on such data. To investigate this, we conduct extensive cross-dataset evaluations between P^{2}V and ITW. Given the class imbalance differences between P^{2}V and ITW (with ITW having a higher proportion of authentic samples), we report comprehensive class-specific performance metrics including precision, recall, and F1-scores for both authentic and deepfake classes.

4.1.2 Detection Models. Following Kawa et al. [16], we adopt two types of audio deepfake detection pipelines. The first is an end-to-end architecture, where the model directly processes raw audio

input and is trained in a fully end-to-end manner. This category includes RawNet3 [15]. The second type is a feature-based pipeline, where audio features are first extracted (Linear Frequency Cepstral Coefficients (LFCC) [11], Mel-Frequency Cepstral Coefficients (MFCC) [32], and the Whisper features [30]) and then passed to a classifier for analysis. Representative models in this category include LCNN [42], MesoNet [1], and SpecRNet [17].

These 4 model architectures and 3 feature types (with their 7 combinations) are used to train a set of 22 baseline audio deepfake detectors. These models output a binary prediction indicating the likelihood of the audio being a deepfake. Models are trained using the binary cross-entropy loss, defined as:

$$\mathcal{L} = -(y \cdot log(p) + (1-y)log(1-p)) \quad (1)$$

where y in Equation 1 denotes the ground-truth label of the input audio and p is the model's predicted probability.

Following the training settings in [16], we use the Adam optimizer across all models. For RawNet3, the learning rate is set to $1e-3$ with a weight decay of $5e-5$. LCNN, MesoNet, and SpecRNet are trained using a learning rate of $1e-4$ and a weight decay of $1e-4$. We use a batch size of 16 and evaluate model performance in 5 epochs, as the model converges. Each experiment is repeated using random seeds 0, 1, and 2. All experiments are conducted using three NVIDIA RTX A6000 GPUs.

For brevity and clarity, results presented in this section represent ensemble predictions averaged across the 22 state-of-the-art baseline models. Detailed performance metrics for individual baselines on P^{2}V[o], P^{2}V[p] and ITW are reported in Tables 2, 4 and 8 in the Appendix.

4.1.3 Evaluation Metrics. The evaluation metrics include precision, recall, and F1 scores for each class individually. We also compute overall precision and recall using weighted averages, where the number of instances per class determines the weights. In addition, we report AUC and Equal Error Rate (EER), widely used metrics in deepfake detection that capture the balance point between false acceptance and false rejection rates. For ease of interpretation, all reported metrics (except for EER) are scaled by a factor of 100.

4.2 Results

4.2.1 Cross-Dataset Evaluations. We assessed the generalization ability of models trained on P^{2}V—both in its original P^{2}V[o] and perturbed P^{2}V[p] form—as well as on the ITW dataset. Figure 2 reports performance across multiple metrics, averaging predictions across 22 baseline models, with rows indicating the test dataset and columns the training dataset. Results for precision and recall are reported in the Appendix.

Across all configurations, models performed best when evaluated on data from the same distribution as their training set. The ITW-trained models achieved near-perfect performance when tested on ITW (99.87% accuracy, 100.00% AUC, 0.00 EER), but their performance dropped dramatically when applied to either P^{2}V[o] (10.14% accuracy, 83.00% AUC, 0.25 EER) or P^{2}V[p] (16.06% accuracy, 84.54% AUC, 0.24 EER). This demonstrates limited out-of-domain transfer and a high degree of dataset specificity. This collapse is particularly evident in the F1 scores: ITW-trained models achieve only 11.55% and 21.92% F1 (Fake) on P^{2}V[o] and P^{2}V[p] respectively, compared to 99.83% on ITW itself. The F1 (Real) scores tell a similar story, with ITW-trained models achieving just 8.69% and 9.25% on P^{2}V[o] and P^{2}V[p], versus 99.90% on ITW. Interestingly, ITW-trained models performed slightly better on P^{2}V[p] than on P^{2}V[o] across most metrics. This counterintuitive result likely occurs because ITW's real-world audio naturally contains noise patterns similar to those we artificially introduced in our perturbed dataset.

In contrast, models trained on P^{2}V variants showed markedly improved cross-domain robustness. Models trained on P^{2}V[o] generalized well to ITW (77.50% accuracy, 97.71% AUC, 0.07 EER, 76.57% F1 Fake), and models trained on P^{2}V[p] maintained comparable performance on ITW (64.36% accuracy, 96.02% AUC, 0.10 EER, 67.53% F1 Fake). Notably, P^{2}V[p]-trained models generalized effectively to P^{2}V[o] (99.15% accuracy, 99.91% AUC, 0.01 EER), performing nearly as well as models trained directly on P^{2}V[o] (99.41% accuracy, 99.93% AUC, 0.01 EER). While models trained on P^{2}V[o] achieved higher accuracy on ITW than P^{2}V[p]-trained models (77.50% vs. 64.36%), the AUC and EER differences were smaller (97.71% vs. 96.02% AUC; 0.07 vs. 0.10 EER), and were not statistically significant (paired t-test, $p = 0.39$ and $p = 0.48$, respectively).

The stark performance asymmetry—where P^{2}V-trained models transfer reasonably to ITW but ITW-trained models catastrophically fail on P^{2}V—highlights the tendency of current detection systems to overfit to dataset-specific characteristics rather than learning generalizable synthetic speech patterns. The stronger cross-domain performance of P^{2}V-trained models suggests that our dataset's varied acoustic conditions and realistic synthesis methods enable more robust real-world deployment.

4.2.2 Evaluation on Recent Real-World Deepfakes. To validate that P^{2}V generalizes beyond benchmark settings, we evaluate detection models on real-world deepfake audio collected in the wild. We have collected a dataset of 255 real-world deepfake audio samples (2023–2024). These samples were collected via three independent channels: (1) Google Alerts monitored continuously since February 2024; (2) eight accredited fact-checking organizations spanning multiple continents—AFP Fact Check, PolitiFact, Snopes, Africa Check, Boomlive, FactCheck.org, OpenSecrets, and the AI Incident Database; and (3) GODDS [29], a journalist-facing detection platform we have deployed since July 2024, through which more than 70 journalists from outlets including *The New York Times*, *The Wall Street Journal*, CNN, and Agence France-Presse (AFP) submitted suspected deepfakes during active investigations. We paired these deepfakes with 255 authentic audio samples from publicly available speeches of well-known deceased public figures in YouTube, creating a balanced evaluation set of 510 samples.

Table 3 presents the performance of four backbone architectures trained on P^{2}V[p] versus ITW. Models trained on P^{2}V consistently outperform ITW-trained models across three of four architectures. LCNN achieves the largest improvement, with AUC increasing from 58.71% (ITW) to 81.95% (P^{2}V) and EER decreasing from 0.420 to 0.259. SpecRNet shows similar gains (AUC: 60.42% → 81.87%, EER: 0.444 → 0.253). MesoNet improves from 54.98% to 67.24% AUC. RawNet3 represents an exception where ITW-trained models perform better (69.29% vs. 56.20% AUC), potentially due to its end-to-end architecture being more sensitive to domain-specific acoustic characteristics present in ITW's web-sourced audio.

Table 2: Performance comparison of baselines on P²V[o]with different feature subsets shown in parentheses. The DDS column is the average of F1-score on the fake class, AUC, and (1-EER).

	Real			Fake			Overall					
Model	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	AUC (↑)	EER (↓)	DDS (↑)
rawnet3	84.70	89.36	86.94	99.52	99.28	99.40	98.89	98.85	98.87	99.22	0.03	98.60
(lfcc)-lcnn	79.23	71.58	74.85	98.74	99.13	98.93	97.90	97.95	97.90	98.84	0.05	97.70
(mfcc)-lcnn	80.80	76.44	78.54	98.95	99.18	99.07	98.17	98.21	98.19	98.04	0.08	96.49
(lfcc-mfcc)-lcnn	75.81	73.50	74.49	98.82	98.93	98.87	97.83	97.84	97.83	96.73	0.10	95.28
(whisper)-lcnn	92.68	83.81	87.93	99.28	99.69	99.48	99.00	99.01	98.99	97.58	0.05	97.24
(whisper-lfcc)-lcnn	91.75	87.76	89.63	99.45	99.64	99.55	99.12	99.13	99.12	99.19	0.03	98.67
(whisper-mfcc)-lcnn	91.66	87.16	89.19	99.43	99.63	99.53	99.09	99.10	99.09	98.85	0.04	98.27
(whisper-lfcc-mfcc)-lcnn	90.89	85.85	88.20	99.37	99.61	99.49	99.01	99.02	99.01	99.01	0.04	98.24
(lfcc)-mesonet	5.57	99.62	10.54	99.95	22.46	34.55	95.91	25.77	33.52	93.03	0.12	71.79
(mfcc)-mesonet	54.77	18.01	11.08	96.35	95.35	95.71	94.57	92.04	92.09	86.88	0.22	86.83
(lfcc-mfcc)-mesonet	7.32	88.56	13.39	99.21	41.91	52.04	95.27	43.90	50.38	87.77	0.21	72.90
(whisper)-mesonet	11.72	64.41	18.26	98.46	65.47	67.93	94.75	65.43	65.80	95.08	0.11	84.15
(whisper-lfcc)-mesonet	38.88	61.39	41.59	98.26	92.48	95.06	95.71	91.15	92.77	93.05	0.14	91.40
(whisper-mfcc)-mesonet	64.91	48.73	31.78	97.66	88.30	91.80	96.25	86.61	89.23	90.60	0.17	88.57
(whisper-lfcc-mfcc)-mesonet	17.36	57.17	26.61	98.09	92.58	95.12	94.64	91.06	92.19	89.25	0.17	89.27
(lfcc)-specrnet	87.94	81.50	84.39	99.18	99.49	99.33	98.69	98.72	98.69	99.15	0.04	98.05
(mfcc)-specrnet	82.71	75.03	78.34	98.89	99.27	99.08	98.20	98.24	98.19	98.15	0.07	96.78
(lfcc-mfcc)-specrnet	81.68	79.59	79.86	99.09	99.16	99.13	98.35	98.33	98.30	97.86	0.06	97.00
(whisper)-specrnet	79.38	89.41	83.94	99.52	98.93	99.22	98.66	98.52	98.57	98.80	0.05	97.72
(whisper-lfcc)-specrnet	94.57	89.05	91.56	99.51	99.75	99.63	99.30	99.30	99.29	99.58	0.03	98.83
(whisper-mfcc)-specrnet	91.63	89.89	90.75	99.55	99.63	99.59	99.21	99.21	99.21	99.39	0.03	98.54
(whisper-lfcc-mfcc)-specrnet	97.06	87.06	91.78	99.42	99.88	99.65	99.32	99.33	99.31	99.43	0.03	98.68

Table 3: Evaluation on Recent Real-World Deepfakes. Performance comparison of detection models trained on P²V[p] versus ITW, evaluated on a dataset containing 255 real-world deepfakes paired with 255 authentic samples. Bold indicates best performance per model. Arrows indicate whether higher (↑) or lower (↓) values are better.

	P²V[p]		ITW	
Model	AUC (↑)	EER (↓)	AUC (↑)	EER (↓)
RawNet3	56.20	0.444	**69.29**	**0.339**
LCNN	**81.95**	**0.259**	58.71	0.420
MesoNet	**67.24**	**0.364**	54.98	0.426
SpecRNet	**81.87**	**0.253**	60.42	0.444

These results demonstrate that P²V-trained models transfer effectively to authentic malicious deepfakes. Crucially, the best P²V-trained detector (LCNN: 81.95% AUC) outperforms the best ITW-trained detector (RawNet3: 69.29% AUC) by over 12 percentage points, confirming that P²V's varied acoustic conditions and modern synthesis methods better prepare models for real-world attacks.

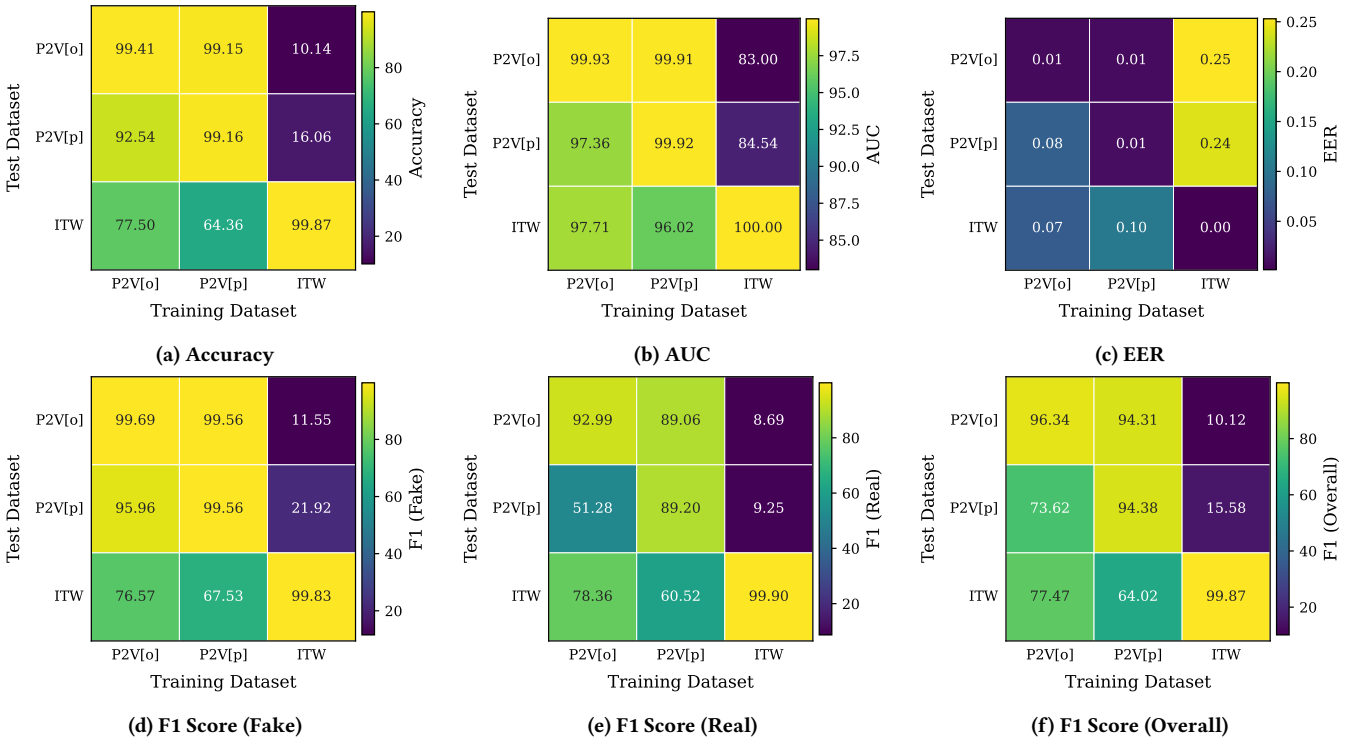
4.2.3 Impact of TTS and VC Methods. We evaluate the performance of predictions averaged across 22 deepfake detection models trained on the ITW dataset across voice samples generated by TTS and VC methods. Table 5 summarizes the results, where models are tested on P²V[p]. To capture overall performance concisely, we also report a Deepfake Detection Score (DDS), calculated as the average of three metrics: F1 score on deepfake audio, AUC, and 1-EER.

The baseline performance (computed on the full P²V[p] test set) achieves AUC of 84.54% and DDS of 60.82%. However, performance varies significantly across different voice cloning methods. We observe DDS scores ranging from 50.52% (XTTS-v2) to 70.67% (Zenos), indicating substantial differences in detection difficulty.

We identify a clear temporal pattern: older voice cloning methods exhibit higher detectability compared to recent approaches. Methods published in 2020-2022 achieve high DDS scores (Tacotron2 DCA: 67.40%, YourTTS: 70.37%), while most methods from 2024-2025 show reduced detectability (XTTS-v2: 50.52%, Seed-VC: 54.05%, VEVO: 52.52%, IndexTTS: 52.52%). Zenos (70.67% DDS score) is the most detectable method, potentially due to artifacts in their generation processes that current detectors can exploit. Performance degradation on newer methods indicates limited generalization of detection models trained on older synthesis techniques, highlighting the need for adaptive detection strategies that can handle new generation capabilities.

To statistically validate performance differences, we conduct Mann-Whitney U tests with Benjamini-Hochberg False Discovery Rate (FDR) correction for multiple hypothesis testing. For each sample, we calculate the detection error as the absolute difference between the averaged detector prediction (y_p) and the ground truth label (y). Each voice cloning method's distribution of these detection errors is compared against all others. All pairwise comparisons yield statistically significant differences ($p < 0.001$ after FDR correction).

4.2.4 Impact of Transcript Generation. P²V incorporates transcripts generated by three different Large Language Models: Gemma-3 (27B parameters) [36], Qwen2.5 (14B parameters) [48], and Mistral



(a) Accuracy **(b) AUC** **(c) EER**

(d) F1 Score (Fake) **(e) F1 Score (Real)** **(f) F1 Score (Overall)**

Figure 2: Cross-dataset generalization performance across audio deepfake detection datasets. Each heatmap shows performance for models trained on the dataset indicated on the x-axis and evaluated on the dataset indicated on the y-axis, averaged across 22 baseline models. Metrics include: (a) accuracy, (b) AUC, (c) EER (lower is better), (d) F1 score for fake class, (e) F1 score for real class, and (f) overall F1 score. Results for precision and recall are reported in the Appendix.

(24B parameters)[7]. We investigate whether these LLMs introduce artifacts that help deepfake detection systems.

Table 6 presents the performance comparison of 22 deepfake detection models across samples containing transcripts generated by each LLM. The results show minimal variation in detection performance: Gemma-3 achieves an Avg score of 60.60%, Qwen2.5 reaches 61.40%, and Mistral attains 60.49%. All three methods perform within a narrow range around the baseline (60.82%).

To validate the statistical significance of observed differences, we conduct Mann-Whitney U tests with Benjamini-Hochberg FDR correction for multiple hypothesis testing. Each LLM's detection error distribution is compared against all others. The analysis reveals no statistically significant differences between transcript generation methods (Gemma-3: $p = 0.48$, Mistral: $p = 0.48$, Qwen2.5: $p = 0.87$ after FDR correction). These findings indicate that current detection systems are not exploiting LLM-specific artifacts in transcript generation, and that transcript quality variations between these models do not create detectable patterns that influence voice cloning detection accuracy.

4.2.5 Impact of Noise Perturbations. Not only can malicious actors employ novel audio generation methods, but they can also use post-processing techniques to evade detection systems. We analyze the vulnerability of deepfake detectors to ten common audio perturbations that adversaries can easily apply to synthetic speech.

Table 7 presents the detection performance when samples are filtered by perturbation type, revealing significant variations in evasion effectiveness. These results demonstrate a clear dichotomy in perturbation impact. Signal processing methods that preserve fundamental audio characteristics show minimal performance differences: Audio Reversal (62.90%) and Impulse Response Application (56.53%) remain close to the baseline (60.82%). Similarly, traditional audio effects like Pitch Shifting (60.45%) and Clipping/Distortion (53.48%) show moderate impact on detection performance.

Conversely, frequency-domain and compression-based perturbations exhibit substantial effects on detection performance. Band-Pass Filtering achieves unexpectedly high detection accuracy (85.08%), suggesting that frequency filtering may actually enhance detection by removing noise while preserving discriminative synthetic artifacts. MP3 Compression (74.53%) similarly improves performance, likely due to the web-sourced nature of the InTheWild training dataset which contains compressed audio samples.

The most effective adversarial manipulations are Background Noise Addition (44.92%) and Gaussian Noise Injection (46.82%), providing attackers with approximately 15-point F1-score reductions from baseline. These techniques achieve extremely low fake detection F1-scores (2.75% and 0.75%, respectively) and high EERs

[7]https://huggingface.co/mistralai/Mistral-Small-24B-Instruct-2501

Table 4: Performance comparison of baselines on P²V[p]with different feature subsets shown in parentheses. The DDS column is the average of F1-score on the fake class, AUC, and (1-EER).

	Real			Fake			Overall					
Model	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	AUC (↑)	EER (↓)	DDS (↑)
rawnet3	90.28	66.37	76.50	98.51	99.68	99.09	98.16	98.25	98.13	96.56	0.09	95.65
(lfcc)-lcnn	89.03	58.64	70.55	98.18	99.68	98.92	97.79	97.92	97.71	97.31	0.08	96.00
(mfcc)-lcnn	88.62	59.51	70.97	98.22	99.66	98.93	97.81	97.94	97.73	96.81	0.09	95.59
(lfcc-mfcc)-lcnn	79.33	68.87	73.70	98.62	99.19	98.90	97.79	97.89	97.82	97.32	0.09	95.80
(whisper)-lcnn	93.95	87.78	90.75	99.45	99.75	99.60	99.22	99.23	99.22	97.47	0.05	97.42
(whisper-lfcc)-lcnn	96.69	85.87	90.93	99.37	99.87	99.62	99.26	99.27	99.25	99.37	0.02	98.84
(whisper-mfcc)-lcnn	98.18	82.65	89.71	99.23	99.93	99.58	99.18	99.19	99.16	98.93	0.03	98.49
(whisper-lfcc-mfcc)-lcnn	96.93	82.50	89.09	99.22	99.88	99.55	99.12	99.14	99.10	98.87	0.04	98.25
(lfcc)-mesonet	29.63	64.41	32.89	98.32	87.50	92.27	95.38	86.51	89.73	91.28	0.17	88.81
(mfcc)-mesonet	60.49	39.10	23.95	97.29	91.61	93.90	95.71	89.36	90.90	89.88	0.18	88.43
(lfcc-mfcc)-mesonet	21.88	68.12	25.70	98.30	76.21	84.83	95.02	75.87	82.30	85.31	0.23	82.23
(whisper)-mesonet	39.19	92.19	46.31	98.94	64.49	66.91	96.38	65.68	66.03	95.81	0.08	84.89
(whisper-lfcc)-mesonet	56.28	63.35	43.35	98.22	76.70	81.52	96.42	76.13	79.88	87.28	0.21	82.66
(whisper-mfcc)-mesonet	40.88	67.55	44.48	98.50	91.46	94.61	96.03	90.44	92.47	92.86	0.15	90.72
(whisper-lfcc-mfcc)-mesonet	90.62	28.42	36.74	96.90	99.73	98.29	96.63	96.68	95.66	89.69	0.17	90.33
(lfcc)-specrnet	87.82	63.77	73.60	98.40	99.60	98.99	97.95	98.06	97.91	97.87	0.07	96.63
(mfcc)-specrnet	87.91	61.52	72.17	98.30	99.59	98.94	97.86	97.96	97.80	97.88	0.08	96.42
(lfcc-mfcc)-specrnet	92.84	57.70	70.83	98.14	99.78	98.95	97.91	97.98	97.75	97.02	0.09	95.65
(whisper)-specrnet	95.33	82.19	88.14	99.21	99.82	99.51	99.04	99.06	99.02	98.81	0.04	98.06
(whisper-lfcc)-specrnet	94.20	89.00	91.48	99.51	99.75	99.63	99.28	99.29	99.28	99.24	0.03	98.71
(whisper-mfcc)-specrnet	96.77	84.41	90.07	99.31	99.87	99.59	99.20	99.21	99.18	98.27	0.04	97.97
(whisper-lfcc-mfcc)-specrnet	87.87	90.13	88.85	99.56	99.43	99.49	99.06	99.03	99.04	99.06	0.03	98.41

Table 5: Impact of TTS and VC Methods. Performance comparison of predictions averaged across 22 deepfake detection models trained on ITW and evaluated on audio generation methods contained in P²V[p]. Each row presents results obtained by filtering the test data to include only samples from the corresponding audio generation method. Metrics include precision (P), recall (R), and F1-score for both real and fake audio detection, along with overall AUC, Equal Error Rate (EER), and Deepfake Detection Score (DDS). Methods are ordered by publication year. Green shading in the DDS column indicates performance above baseline (60.82%), while red shading indicates performance below baseline. Arrows indicate whether higher (↑) or lower (↓) values are better.

	Real			Fake			Overall					
Audio Generation Model	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	AUC (↑)	EER (↓)	DDS (↑)
Baseline	4.85	99.90	9.25	99.96	12.31	21.92	52.41	56.11	15.58	84.54	0.240	60.82
Tacotron2 DCA [2]	33.59	99.90	50.28	99.62	11.64	20.85	66.61	55.77	35.56	94.49	0.131	67.40
YourTTS [6]	36.12	99.90	53.06	99.79	20.96	34.65	67.96	60.43	43.85	92.24	0.158	70.37
Diff-HierVC [10]	34.42	99.90	51.20	99.71	14.85	25.84	67.06	57.37	38.52	85.14	0.234	62.54
FreeVC [20]	32.95	99.90	49.55	99.52	9.04	16.57	66.23	54.47	33.06	91.58	0.175	63.55
CosyVoice2 [13]	34.13	99.90	50.88	99.68	13.75	24.17	66.91	56.83	37.52	81.53	0.272	59.51
XTTS-v2 [5]	32.01	99.90	48.49	99.15	5.09	9.68	65.58	52.49	29.08	74.11	0.322	50.52
Seed-VC [21]	32.90	99.90	49.50	99.51	8.84	16.23	66.20	54.37	32.86	76.38	0.305	54.05
Zonos (2025)	35.84	99.90	52.76	99.78	20.00	33.32	67.81	59.95	43.04	93.11	0.144	70.67
IndexTTS [12]	32.70	99.90	49.27	99.46	8.00	14.82	66.08	53.95	32.04	74.79	0.320	52.52
VEVO [57]	33.41	99.90	50.07	99.60	10.92	19.68	66.50	55.41	34.88	82.06	0.262	58.51

(0.363 and 0.331, respectively), demonstrating that readily accessible and human-imperceptible noise injection can severely compromise detection systems.

To validate observed performance differences, we conduct Mann-Whitney U tests with Benjamini-Hochberg False Discovery Rate (FDR) correction for multiple hypothesis testing. Each noise perturbation method's detection error distribution is compared against all others. All pairwise comparisons yield statistically significant differences ($p < 0.001$ after FDR correction), confirming that performance variations are not due to random variation.

5 Limitations

We acknowledge several limitations. First, P²V focuses exclusively on English-speaking deceased public figures, which limits direct

Table 6: Impact of Transcript Generation Methods. Performance comparison of predictions averaged across 22 deepfake detection models trained on ITW and evaluated on transcript generation methods (LLMs) contained in P^{2}V[p]. Each row presents results obtained by filtering the test data to include only samples generated with the corresponding LLM. Metrics include precision (P), recall (R), and F1-score for both real and fake audio detection, along with overall AUC, Equal Error Rate (EER), and Deepfake Detection Score (DDS). Green shading in the DDS column indicates performance above baseline (60.82%), while red shading indicates performance below baseline. Arrows indicate whether higher (↑) or lower (↓) values are better.

	Real			Fake			Overall					
LLM	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	AUC (↑)	EER (↓)	DDS (↑)
Baseline	4.85	99.90	9.25	99.96	12.31	21.92	52.41	56.11	15.58	84.54	0.240	60.82
Gemma-3 (27B)	13.28	99.90	23.44	99.89	12.45	22.14	56.59	56.17	22.79	84.03	0.244	60.60
Qwen2.5 (14B)	13.27	99.90	23.43	99.89	12.39	22.05	56.58	56.15	22.74	85.35	0.232	61.40
Mistral (24B)	13.23	99.90	23.37	99.89	12.09	21.56	56.56	55.99	22.47	84.25	0.243	60.49

Table 7: Impact of Noise Perturbations. Performance comparison of predictions averaged across 22 deepfake detection models trained on ITW and evaluated on noise perturbations contained in P^{2}V[p]. Each row presents results obtained by filtering the test data to include only samples generated with the corresponding perturbation. Metrics include precision (P), recall (R), and F1-score for both real and fake audio detection, along with overall AUC, Equal Error Rate (EER), and Deepfake Detection Score (DDS). Green shading in the DDS column indicates performance above baseline (60.82%), while red shading indicates performance below baseline. Arrows indicate whether higher (↑) or lower (↓) values are better.

	Real			Fake			Overall					
Noise Perturbation	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	AUC (↑)	EER (↓)	DDS (↑)
Baseline	4.85	99.90	9.25	99.96	12.31	21.92	52.41	56.11	15.58	84.54	0.240	60.82
Audio Reversal	32.33	99.90	48.86	99.35	6.69	12.54	65.84	53.30	30.70	92.09	0.159	62.90
Background Noise Addition	30.68	99.90	46.94	96.88	1.33	2.62	63.78	50.62	24.78	68.45	0.363	44.92
Impulse Response Application	32.23	99.90	48.73	99.24	5.74	10.85	65.73	52.82	29.79	83.45	0.247	56.55
Time Stretching	32.97	99.90	49.58	99.46	8.18	15.11	66.22	54.04	32.35	84.01	0.248	58.11
ClippingDistortion	31.54	99.90	47.95	97.92	2.08	4.08	64.73	50.99	26.01	81.98	0.256	53.48
Pitch Shifting	33.26	99.90	49.91	99.56	9.91	18.03	66.41	54.91	33.97	86.02	0.227	60.45
Band-Pass Filtering	48.35	99.90	65.16	99.92	52.43	68.77	74.13	76.16	66.97	96.59	0.101	85.08
Air Absorption Simulation	33.03	99.90	49.64	99.52	9.16	16.78	66.27	54.53	33.21	85.81	0.227	59.95
MP3 Compression	37.80	99.90	54.85	99.84	26.97	42.46	68.82	63.43	48.66	94.36	0.132	74.53
Gaussian Noise Injection	31.11	99.90	47.45	89.47	0.38	0.75	60.29	50.14	24.10	72.82	0.331	46.82

multilingual applicability but ensures IRB compliance and methodological consistency. The underlying pipeline is language-agnostic and extensible to other languages with appropriate TTS models. Second, the dataset exhibits class imbalance, which is common among deepfake benchmarks [14, 37, 44]. We address this by reporting threshold-independent metrics (AUC, EER) alongside class-specific scores, and our real-world evaluation demonstrates that P^{2}V-trained models generalize effectively despite this imbalance. Finally, while P^{2}V provides rich metadata enabling attribution analysis, future work could leverage this to develop interpretability tools explaining detector decisions.

6 Conclusion

Audio deepfake detection faces a critical trust gap: current systems excel on artificial benchmarks but fail against real-world challenges. In light of this, we have proposed a novel large-scale and IRB-approved dataset, namely Perturbed Public Voices (P^{2}V), consisting of 257,440 samples from 206 deceased public figures and incorporating news-guided transcripts as well as stateof-the-art audio generation techniques and perturbations that can be encountered in the real world. We have demonstrated that adversarial perturbations and evolving synthesis methods systematically degrade state-of-the-art detectors performance, exposing vulnerabilities that existing datasets overlook. Our results establish that robustness requires training on data that mirrors the complexity of malicious deepfakes, including noise, plausible content, and state-of-the-art voice cloning. P^{2}V provides this foundation, enabling models that maintain reliability against attacks while generalizing to in-the-wild scenarios.

References

[1] Darius Afchar, Vincent Nozick, Junichi Yamagishi, and Isao Echizen. 2018. Mesonet: a compact facial video forgery detection network. In *2018 IEEE international workshop on information forensics and security (WIFS)*. IEEE, 1–7.
[2] Eric Battenberg, RJ Skerry-Ryan, Soroosh Mariooryad, Daisy Stanton, David Kao, Matt Shannon, and Tom Bagby. 2020. Location-relative attention mechanisms for robust long-form speech synthesis. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 6194–6198.
[3] Jordan J Bird and Ahmad Lotfi. 2023. Real-time Detection of AI-Generated Speech for DeepFake Voice Conversion. *arXiv preprint arXiv:2308.12734* (2023).

[4] Hervé Bredin. 2023. pyannote.audio 2.1 speaker diarization pipeline: principle, benchmark, and recipe. In *Proc. INTERSPEECH 2023*.

[5] Edresson Casanova, Kelly Davis, Eren Gölge, Görkem Göknar, Iulian Gulea, Logan Hart, Aya Aljafari, Joshua Meyer, Reuben Morais, Samuel Olayemi, et al. 2024. XTTS: a Massively Multilingual Zero-Shot Text-to-Speech Model. In *Proc. Interspeech 2024*. 4978–4982.

[6] Edresson Casanova, Julian Weber, Christopher D Shulby, Arnaldo Candido Junior, Eren Gölge, and Moacir A Ponti. 2022. Yourtts: Towards zero-shot multi-speaker tts and zero-shot voice conversion for everyone. In *International conference on machine learning*. PMLR, 2709–2720.

[7] Nuo Chen, Yan Wang, Yang Deng, and Jia Li. 2024. The oscars of ai theater: A survey on role-playing with language models. *arXiv preprint arXiv:2407.11484* (2024).

[8] Yujie Chen, Jiangyan Yi, Cunhang Fan, Jianhua Tao, Yong Ren, Siding Zeng, Chu Yuan Zhang, Xinrui Yan, Hao Gu, Jun Xue, Chenglong Wang, Zhao Lv, and Xiaohui Zhang. 2025. Region-Based Optimization in Continual Learning for Audio Deepfake Detection. In *AAAI-25, Sponsored by the Association for the Advancement of Artificial Intelligence, February 25 - March 4, 2025, Philadelphia, PA, USA*, Toby Walsh, Julie Shah, and Zico Kolter (Eds.). AAAI Press, 23651–23659. doi:10.1609/AAAI.V39I22.34535

[9] Ziyang Chen and Stylios Moscholios. 2024. Using Prompts to Guide Large Language Models in Imitating a Real Person's Language Style. *arXiv preprint arXiv:2410.03848* (2024).

[10] Ha Yeong Choi, Sang Hoon Lee, and Seong Whan Lee. 2023. Diff-HierVC: Diffusion-based Hierarchical Voice Conversion with Robust Pitch Generation and Masked Prior for Zero-shot Speaker Adaptation. In *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*, Vol. 2023. 2283–2287.

[11] Steven Davis and Paul Mermelstein. 1980. Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. *IEEE transactions on acoustics, speech, and signal processing* 28, 4 (1980), 357–366.

[12] Wei Deng, Siyi Zhou, Jingchen Shu, Jinchao Wang, and Lu Wang. 2025. IndexTTS: An Industrial-Level Controllable and Efficient Zero-Shot Text-To-Speech System. *arXiv preprint arXiv:2502.05512* (2025).

[13] Zhihao Du, Yuxuan Wang, Qian Chen, Xian Shi, Xiang Lv, Tianyu Zhao, Zhifu Gao, Yexin Yang, Changfeng Gao, Hui Wang, et al. 2024. Cosyvoice 2: Scalable streaming speech synthesis with large language models. *arXiv preprint arXiv:2412.10117* (2024).

[14] Joel Frank and Lea Schönherr. [n. d.]. WaveFake: A Data Set to Facilitate Audio Deepfake Detection. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*.

[15] Jee-weon Jung, You Jin Kim, Hee-Soo Heo, Bong-Jin Lee, Youngki Kwon, and Joon Son Chung. 2022. Pushing the limits of raw waveform speaker recognition. *arXiv preprint arXiv:2203.08488* (2022).

[16] Piotr Kawa, Marcin Plata, Michał Czuba, Piotr Szymański, and Piotr Syga. 2023. Improved deepfake detection using whisper features. *arXiv preprint arXiv:2306.01428* (2023).

[17] Piotr Kawa, Marcin Plata, and Piotr Syga. 2022. Specrnet: Towards faster and more accessible audio deepfake detection. In *2022 IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*. IEEE, 792–799.

[18] Tomi Kinnunen, Md Sahidullah, Héctor Delgado, Massimiliano Todisco, Nicholas Evans, Junichi Yamagishi, and Kong Aik Lee. 2017. The ASVspoof 2017 challenge: Assessing the limits of replay spoofing attack detection. In *Interspeech 2017*. International Speech Communication Association, 2–6.

[19] Tomi Kinnunen, Zhizheng Wu, EURECOM Nicholas Evans, and Junichi Yamagishi. 2018. Automatic Speaker Verification Spoofing and Countermeasures Challenge (ASVspoof 2015) Database. (2018).

[20] Jingyi Li, Weiping Tu, and Li Xiao. 2023. Freevc: Towards high-quality text-free one-shot voice conversion. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1–5.

[21] Songting Liu. 2024. Zero-shot Voice Conversion with Diffusion Transformers. *arXiv preprint arXiv:2411.09943* (2024).

[22] Keming Lu, Bowen Yu, Chang Zhou, and Jingren Zhou. 2024. Large Language Models are Superpositions of All Characters: Attaining Arbitrary Role-play via Self-Alignment. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 7828–7840.

[23] Haoxin Ma, Jiangyan Yi, Chenglong Wang, Xinrui Yan, Jianhua Tao, Tao Wang, Shiming Wang, and Ruibo Fu. 2024. CFAD: A Chinese dataset for fake audio detection. *Speech Communication* 164 (2024), 103122.

[24] Nicolas M Müller, Pavel Czempin, Franziska Dieckmann, Adam Froghyar, and Konstantin Böttinger. 2022. Does audio deepfake detection generalize? *arXiv preprint arXiv:2203.16263* (2022).

[25] Nicolas M Müller, Piotr Kawa, Wei Herng Choong, Edresson Casanova, Eren Gölge, Thorsten Müller, Piotr Syga, Philip Sperl, and Konstantin Böttinger. 2024. Mlaad: The multi-language audio anti-spoofing dataset. *arXiv preprint arXiv:2401.09512* (2024).

[26] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. *Advances in neural information processing systems* 35 (2022), 27730–27744.

[27] Karol J Piczak. 2015. ESC: Dataset for environmental sound classification. In *Proceedings of the 23rd ACM international conference on Multimedia*. 1015–1018.

[28] Alexis Plaquet and Hervé Bredin. 2023. Powerset multi-class cross entropy loss for neural speaker diarization. In *Proc. INTERSPEECH 2023*.

[29] Marco Postiglione, Julian Baldwin, Natalia Denisenko, Luke Fosdick, Chongyang Gao, Isabel Gortner, Chiara Pulice, Sarit Kraus, and VS Subrahmanian. 2024. GODDS: The Global Online Deepfake Detection System. In *AAAI 2025 Demonstration Program*. https://openreview.net/forum?id=gGaJbk2o8A

[30] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*. PMLR, 28492–28518.

[31] Margie Ruffin, Haeseung Seo, Aiping Xiong, and Gang Wang. 2024. Does It Matter Who Said It? Exploring the Impact of Deepfake-Enabled Profiles on User Perception towards Disinformation. In *Proceedings of the Eighteenth International AAAI Conference on Web and Social Media, ICWSM 2024, Buffalo, New York, USA, June 3-6, 2024*, Yu-Ru Lin, Yelena Mejova, and Meeyoung Cha (Eds.). AAAI Press, 1328–1341. doi:10.1609/ICWSM.V18I1.31392

[32] Md Sahidullah and Goutam Saha. 2012. Design, analysis and experimental evaluation of block based transformation in MFCC computation for speaker recognition. *Speech communication* 54, 4 (2012), 543–565.

[33] Davide Salvi, Brian Hosler, Paolo Bestagini, Matthew C Stamm, and Stefano Tubaro. 2023. TIMIT-TTS: A text-to-speech dataset for multimodal synthetic media detection. *IEEE access* 11 (2023), 50851–50866.

[34] Murray Shanahan, Kyle McDonell, and Laria Reynolds. 2023. Role play with large language models. *Nature* 623, 7987 (2023), 493–498.

[35] Chengzhe Sun, Shan Jia, Shuwei Hou, and Siwei Lyu. 2023. Ai-synthesized voice detection using neural vocoder artifacts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 904–912.

[36] Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, et al. 2025. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786* (2025).

[37] Massimiliano Todisco, Xin Wang, Ville Vestman, Md Sahidullah, Héctor Delgado, Andreas Nautsch, Junichi Yamagishi, Nicholas Evans, Tomi Kinnunen, and Kong Aik Lee. 2019. ASVspoof 2019: Future horizons in spoofed and fake audio detection. *arXiv preprint arXiv:1904.05441* (2019).

[38] Christina P. Walker, Daniel S. Schiff, and Kaylyn Jackson Schiff. 2024. Merging AI Incidents Research with Political Misinformation Research: Introducing the Political Deepfakes Incidents Database. In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2014, February 20-27, 2024, Vancouver, Canada*, Michael J. Wooldridge, Jennifer G. Dy, and Sriraam Natarajan (Eds.). AAAI Press, 23053–23058. doi:10.1609/AAAI.V38I21.30349

[39] Xin Wang, Héctor Delgado, Hemlata Tak, Jee-weon Jung, Hye-jin Shim, Massimiliano Todisco, Ivan Kukanov, Xuechen Liu, Md Sahidullah, Tomi Kinnunen, et al. 2024. ASVspoof 5: Crowdsourced speech data, deepfakes, and adversarial attacks at scale. *arXiv preprint arXiv:2408.08739* (2024).

[40] Jason Wei, Maarten Bosma, Vincent Y Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M Dai, and Quoc V Le. 2021. Finetuned language models are zero-shot learners. *arXiv preprint arXiv:2109.01652* (2021).

[41] Haolin Wu, Jing Chen, Ruiying Du, Cong Wu, Kun He, Xingcan Shang, Hao Ren, and Guowen Xu. 2024. CLAD: Robust Audio Deepfake Detection Against Manipulation Attacks with Contrastive Learning. *arXiv preprint arXiv:2404.15854* (2024).

[42] Xiang Wu, Ran He, Zhenan Sun, and Tieniu Tan. 2018. A light CNN for deep face representation with noisy labels. *IEEE transactions on information forensics and security* 13, 11 (2018), 2884–2896.

[43] Zhizheng Wu, Junichi Yamagishi, Tomi Kinnunen, Cemal Hanilçi, Mohammed Sahidullah, Aleksandr Sizov, Nicholas Evans, Massimiliano Todisco, and Hector Delgado. 2017. ASVspoof: the automatic speaker verification spoofing and countermeasures challenge. *IEEE Journal of Selected Topics in Signal Processing* 11, 4 (2017), 588–604.

[44] Yuankun Xie, Yi Lu, Ruibo Fu, Zhengqi Wen, Zhiyong Wang, Jianhua Tao, Xin Qi, Xiaopeng Wang, Yukun Liu, Haonan Cheng, et al. 2024. The Codecfake Dataset and Countermeasures for the Universally Detection of Deepfake Audio. *arXiv preprint arXiv:2405.04880* (2024).

[45] Yuankun Xie, Jingjing Zhou, Xiaolin Lu, Zhenghao Jiang, Yuxin Yang, Haonan Cheng, and Long Ye. 2024. FSD: An initial chinese dataset for fake song detection. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 4605–4609.

[46] Junichi Yamagishi, Xin Wang, Massimiliano Todisco, Md Sahidullah, Jose Patino, Andreas Nautsch, Xuechen Liu, Kong Aik Lee, Tomi Kinnunen, Nicholas Evans, et al. 2021. ASVspoof 2021: accelerating progress in spoofed and deepfake speech detection. In *ASVspoof 2021 Workshop-Automatic Speaker Verification and Spoofing*

Coutermeasures Challenge.
[47] Xinrui Yan, Jiangyan Yi, Chenglong Wang, Jianhua Tao, Junzuo Zhou, Hao Gu, and Ruibo Fu. 2022. System Fingerprint Recognition for Deepfake Audio: An Initial Dataset and Investigation. *arXiv preprint arXiv:2208.10489* (2022).
[48] An Yang, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoyan Huang, Jiandong Jiang, Jianhong Tu, Jianwei Zhang, Jingren Zhou, et al. 2025. Qwen2. 5-1M Technical Report. *arXiv preprint arXiv:2501.15383* (2025).
[49] Jiangyan Yi, Ruibo Fu, Jianhua Tao, Shuai Nie, Haoxin Ma, Chenglong Wang, Tao Wang, Zhengkun Tian, Ye Bai, Cunhang Fan, et al. 2022. Add 2022: the first audio deep synthesis detection challenge. In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 9216–9220.
[50] Jiangyan Yi, Jianhua Tao, Ruibo Fu, Xinrui Yan, Chenglong Wang, Tao Wang, Chu Yuan Zhang, Xiaohui Zhang, Yan Zhao, Yong Ren, et al. 2023. Add 2023: the second audio deepfake detection challenge. *arXiv preprint arXiv:2305.13774* (2023).
[51] Jiangyan Yi, Chenglong Wang, Jianhua Tao, Xiaohui Zhang, Chu Yuan Zhang, and Yan Zhao. 2023. Audio deepfake detection: A survey. *arXiv preprint arXiv:2308.14970* (2023).
[52] Hibiki Yokoyama, Rikuto Tsuchida, Kosei Buma, Sho Miyakawa, Takehito Utsuro, and Masaharu Yoshioka. 2024. Aggregating Impressions on Celebrities and their Reasons from Microblog Posts and Web Search Pages. In *Proceedings of the 3rd Workshop on Knowledge Augmented Methods for NLP*. 59–72.
[53] Kuiyuan Zhang, Zhongyun Hua, Rushi Lan, Yushu Zhang, and Yifang Guo. 2025. Phoneme-Level Feature Discrepancies: A Key to Detecting Sophisticated Speech Deepfakes. In *AAAI-25, Sponsored by the Association for the Advancement of Artificial Intelligence, February 25 - March 4, 2025, Philadelphia, PA, USA*, Toby Walsh, Julie Shah, and Zico Kolter (Eds.). AAAI Press, 1066–1074. doi:10.1609/AAAI.V39I1.32093
[54] Lin Zhang, Xin Wang, Erica Cooper, Nicholas Evans, and Junichi Yamagishi. 2022. The partialspoof database and countermeasures for the detection of short fake speech segments embedded in an utterance. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 31 (2022), 813–825.
[55] Lin Zhang, Xin Wang, Erica Cooper, Junichi Yamagishi, Jose Patino, and Nicholas Evans. 2021. An initial investigation for detecting partially spoofed audio. *arXiv preprint arXiv:2104.02518* (2021).
[56] Xiaohui Zhang, Jiangyan Yi, Chenglong Wang, Chu Yuan Zhang, Siding Zeng, and Jianhua Tao. 2024. What to Remember: Self-Adaptive Continual Learning for Audio Deepfake Detection. In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2014, February 20-27, 2024, Vancouver, Canada*, Michael J. Wooldridge, Jennifer G. Dy, and Sriraam Natarajan (Eds.). AAAI Press, 19569–19577. doi:10.1609/AAAI.V38I17.29929
[57] Xueyao Zhang, Xiaohui Zhang, Kainan Peng, Zhenyu Tang, Vimal Manohar, Yingru Liu, Jeff Hwang, Dangna Li, Yuhao Wang, Julian Chan, et al. 2025. Vevo: Controllable zero-shot voice imitation with self-supervised disentanglement. *arXiv preprint arXiv:2502.07243* (2025).
[58] Yan Zhao, Jiangyan Yi, Jianhua Tao, Chenglong Wang, and Yongfeng Dong. 2024. EmoFake: An initial dataset for emotion fake audio detection. In *China National Conference on Chinese Computational Linguistics*. Springer, 419–433.

A Appendix

A.1 Transcript Generation Prompt

This section details the prompt templates used to generate the four types of speech transcripts for our audio deepfake dataset. The prompts are designed to systematically vary two key dimensions: contextual grounding (news-guided vs. non-news-guided) and persona authenticity (in-character vs. out-of-character). The template is provided as follows:

Here, we are including the first section only when producing news-guided transcripts. In such cases, we provide the headline of the retrieved news article (`title_tokens`) and a concise summary of the news article content (`summary_tokens`). `subject_tokens` represents the name of the deceased public figure (DPF) being impersonated, and `not` is a conditional modifier that is included for out-of-character conditions and omitted for in-character conditions.

A.2 Detailed Experiments Results

We presented the detailed results in this section across the 22 state-of-the-art baseline models. We train and test the baseline models on the same dataset. The detailed performance metrics for individual baselines trained and tested on ITW are provided in Tables 8.

A.3 Cross-Dataset Evaluations

Figure 3 reports heatmaps for precision and recall metrics across 22 baseline models, with rows indicating the test dataset and columns the training dataset. Accuracy, AUC, EER, and F1 scores are reported in the main body (Figure 2).

A.4 Computational Cost

All GPU-based stages were run on 3× NVIDIA RTX A6000 GPUs. Dataset construction costs: reference audio collection and diarization required 10 hours of automated processing plus 12 hours of manual verification; transcript generation across the 3 LLMs (82,400 transcripts) required 493.5 cumulative GPU-hours; audio synthesis across the 10 TTS/VC pipelines (247,200 samples) required 29 cumulative GPU-days; and audio perturbation required 8 hours of wall-clock time on a 32-core CPU node. Detector training and inference costs are summarized in Table 9.

Table 9: Computational cost of the four backbone detector architectures.

Model	Params (M)	Train (min/ep)	Inf. (ms/sample)
RawNet3	15.50	501	25.6
LCNN	7.20	57	20.0
MesoNet	0.03	45	6.3
SpecRNet	0.28	46	9.4

Prompt

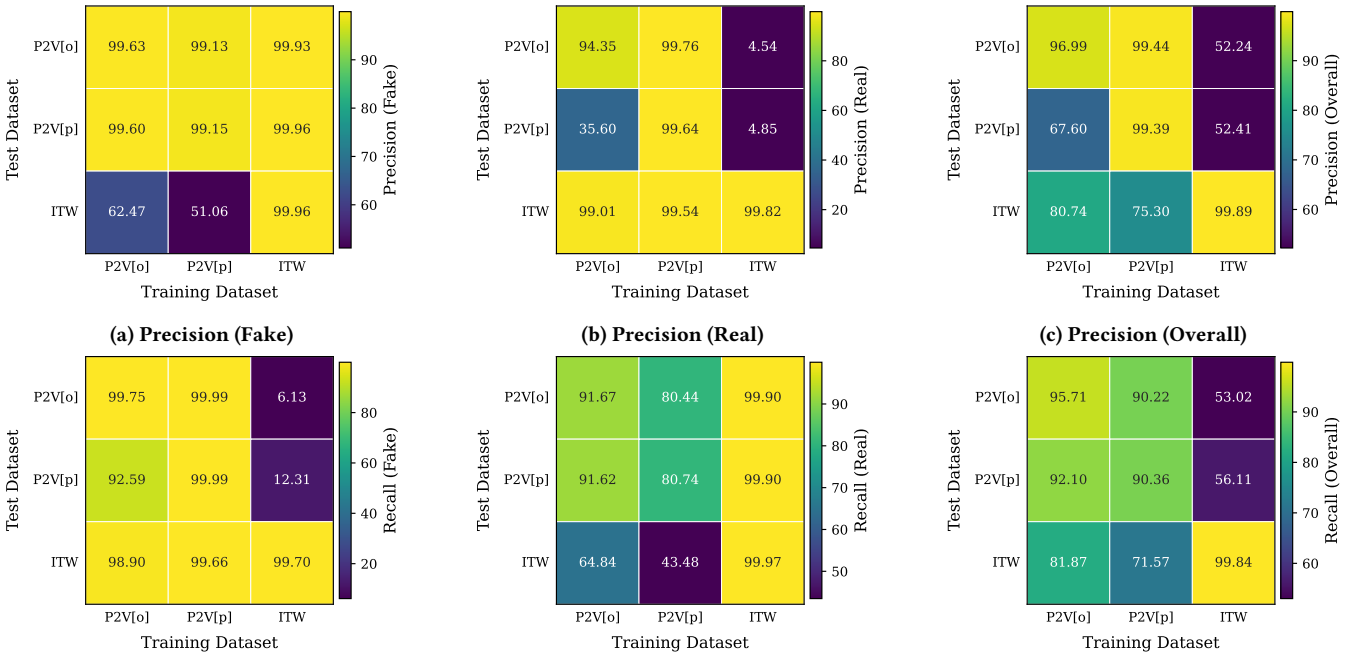
If News-Guided:

```
Consider this news piece along with its summary:
"""
Title: {title_tokens}
Summary: {summary_tokens}
"""
Generate a realistic quote that would {not} likely be said by {subject_tokens},
maintaining their authentic speech patterns and vocabulary. The quote should
be 100 words in length. Only output quote without other words.
```

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Table 8: Performance comparison of the baselines on ITW with different feature subsets shown in parentheses. The Avg column is the average of F1-score on the fake class, AUC, and (1-EER).

	Real			Fake			Overall					
Model	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	P (↑)	R (↑)	F1 (↑)	AUC (↑)	EER (↓)	Avg. (↑)
rawnet3	99.59	99.91	99.75	99.84	99.31	99.58	99.69	99.69	99.69	99.97	0.00	99.72
(lfcc)-lcnn	99.82	99.79	99.81	99.65	99.70	99.68	99.76	99.76	99.76	99.98	0.00	99.82
(mfcc)-lcnn	99.78	99.77	99.78	99.62	99.63	99.63	99.72	99.72	99.72	99.99	0.00	99.77
(lfcc-mfcc)-lcnn	99.73	99.75	99.74	99.58	99.55	99.56	99.68	99.67	99.67	99.99	0.00	99.74
(whisper)-lcnn	92.88	92.45	92.66	87.34	88.02	87.68	90.82	90.80	90.81	96.54	0.10	91.49
(whisper-lfcc)-lcnn	99.87	99.87	99.87	99.77	99.79	99.78	99.84	99.84	99.84	99.99	0.00	99.87
(whisper-mfcc)-lcnn	99.83	99.79	99.81	99.65	99.72	99.68	99.76	99.76	99.76	99.98	0.00	99.81
(whisper-lfcc-mfcc)-lcnn	99.84	99.90	99.87	99.83	99.73	99.78	99.84	99.84	99.84	99.99	0.00	99.85
(lfcc)-mesonet	99.57	99.40	99.48	98.99	99.27	99.13	99.35	99.35	99.35	99.93	0.01	99.46
(mfcc)-mesonet	87.27	99.55	92.33	99.17	69.47	74.91	91.69	88.37	85.86	98.82	0.03	90.23
(lfcc-mfcc)-mesonet	98.73	99.30	99.01	98.81	97.84	98.32	98.76	98.76	98.76	99.86	0.01	98.97
(whisper)-mesonet	83.97	83.35	82.75	75.37	71.11	71.05	80.78	78.80	78.40	87.68	0.20	79.68
(whisper-lfcc)-mesonet	93.72	99.53	96.42	99.20	87.84	92.62	95.76	95.19	95.01	99.71	0.01	96.97
(whisper-mfcc)-mesonet	88.45	99.74	93.31	99.54	74.11	81.31	92.57	90.21	88.84	99.47	0.03	92.64
(whisper-lfcc-mfcc)-mesonet	97.59	98.75	98.13	97.95	95.71	96.72	97.72	97.62	97.60	99.69	0.02	98.28
(lfcc)-specrnet	99.70	99.89	99.80	99.82	99.49	99.65	99.74	99.74	99.74	99.99	0.00	99.80
(mfcc)-specrnet	99.63	99.83	99.73	99.72	99.38	99.55	99.66	99.66	99.66	99.97	0.00	99.70
(lfcc-mfcc)-specrnet	99.82	99.82	99.82	99.69	99.69	99.69	99.77	99.77	99.77	99.99	0.00	99.82
(whisper)-specrnet	93.00	91.83	92.40	86.51	88.29	87.37	90.59	90.51	90.53	96.49	0.10	91.38
(whisper-lfcc)-specrnet	99.74	99.87	99.81	99.79	99.56	99.68	99.76	99.76	99.76	99.98	0.00	99.78
(whisper-mfcc)-specrnet	99.68	99.76	99.72	99.59	99.46	99.53	99.65	99.65	99.65	99.96	0.00	99.70
(whisper-lfcc-mfcc)-specrnet	99.81	99.67	99.74	99.45	99.68	99.56	99.68	99.67	99.67	99.98	0.00	99.74



(a) Precision (Fake) **(b) Precision (Real)** **(c) Precision (Overall)**

(d) Recall (Fake) **(e) Recall (Real)** **(f) Recall (Overall)**

Figure 3: Cross-dataset precision and recall. Each heatmap shows performance for models trained on the x-axis dataset and evaluated on the y-axis dataset, averaged across 22 baselines.